

HORIZONTAL p -ADIC L -FUNCTIONS

ABSTRACT. Talk by Asbjørn Nordentoft at the seminar at Aarhus on 13 May 2024. Joint work with Daniel Kriz.

1. INTRODUCTION

Let E be an elliptic curve over $\mathbb{Q}$. Let L/K be a Galois extension of number fields. Then there is the following concept, due to Mazur and Rubin.

Definition 1. We say that E is *diophantine stable* (abbreviated DS) relative to L/K if

$$\text{rank}_{\mathbb{Z}} E(L) = \text{rank}_{\mathbb{Z}} E(K).$$

Question 2. How often is E diophantine stable for L/K ?

Remark 3. The notion of diophantine stability has applications to Hilbert's tenth problem. This was part of the original motivation. From our point of view, it's just a natural question.

There is an analytic counterpart to this question. Let $F/\mathbb{Q}$ be an abelian extension of $\mathbb{Q}$. Assuming the BSD conjecture, the question of diophantine stability takes a different form:

$$\begin{aligned} \text{rank}_{\mathbb{Z}} E(F) &= \text{ord}_{s=1} L(E/F, s) \\ &= \sum_{\chi \in \widehat{\text{Gal}(F/\mathbb{Q})}} \text{ord}_{s=1} L(E, \chi, s) \\ &= \text{rank}_{\mathbb{Z}} E(\mathbb{Q}) + \sum_{1 \neq \chi \in \widehat{\text{Gal}(F/\mathbb{Q})}} \text{ord}_{s=1} L(E, \chi, s). \end{aligned}$$

where

$$L(E, \chi, s) = \sum_{n \geq 1} a_E(n) \chi(n) n^{-s}.$$

The upshot is that diophantine stability for $F/\mathbb{Q}$ is related, under BSD, to understanding when $L(E, \chi, 1) \neq 0$ for $\chi \in \widehat{\text{Gal}(F/\mathbb{Q})}$ with $\chi \neq 1$.

Question 4. How often is $L(E, \chi, 1) \neq 0$ for Dirichlet characters χ ?

2. VERTICAL ANALYSIS

Let $F_n = \mathbb{Q}(\mu_p)$, $F_{\infty} = \cup_{n \geq 1} F_n$ (p th cyclotomic extension of $\mathbb{Q}$).

Theorem 5 (Mazur). *If E is good at p , then $\text{rank}_{\mathbb{Z}} E(F_{\infty}) < \infty$.*

Remark 6. In our language of Diophantine stability, this means that if n is sufficiently large, then E is diophantine stable for F_{∞}/F_n .

Theorem 7 (Rohrlich). *We have $L(E, \chi, 1) \neq 0$ for all but finitely many χ , a Dirichlet character of p -power conductor. This is an exercise in class field theory.*

Remark 8. The results of Mazur and Rohrlich should be equivalent under BSD, but the speaker doesn't know of a direct way to go between them.

3. HORIZONTAL ANALYSIS

Fix $d \geq 2$. Let $L/\mathbb{Q}$ be a cyclic extension, of order d . (The corresponding Dirichlet characters χ then have order d .)

Conjecture 9 (Goldfeld). *Suppose $d = 2$, and let χ_D be a quadratic character of conductor D . Then*

$$\text{ord}_{s=1} L(E, \chi_D, s) = \begin{cases} 0 & \text{with probability 50\%,} \\ 1 & \text{with probability 50\%,} \\ \geq 2 & \text{with probability 0\%.} \end{cases}$$

Previous work when $d = 2$:

- Friedberg–Hoffstein, 1990's
- Murty–Murty, Ono–Skinner ($\gg X/\log X$);
- Smith–Kriz

How about when $d > 2$:

Conjecture 10 (David–Fearnley–Kisilevski).

4. RESULTS

We denote by $F_d(X)$ the set of relevant Dirichlet characters $\chi \pmod{D}$ with $D \leq X$ of order τ . There exists $c_d > 0$ such that

$$\#F_d(X) \sim c_d X (\log X)^{\sigma_0(d)-2}.$$

Here $\sigma_0(d)$ denotes the number of divisors of d .

Theorem 11 (Kriz–N 23). *Let $d \equiv 2 \pmod{4}$, $d > 2$. Then there exists $\alpha = \alpha(E, d) > 0$ such that*

$$\#\{x \in F_d(X) : L(E, \chi, 1) \neq 0\} \gg \frac{X}{(\log X)^{-\alpha}}.$$

Remark 12. Not previously known in this setting for $n = 6$.

We can also study a more general case, by imposing further assumptions.

Theorem 13. *We get the same conclusions in the following cases:*

- (1) $L(E, 1) \neq 0$ and there exists $p \mid d$ such that $\bar{\rho}_{E,p}$ is irreducible,
- (2) $2 \mid d$ and $\bar{\rho}_{E,2}$ is irreducible.

Remark 14. $d = 2^m, \overline{\rho_{E,2}}$ if and only if $E(\mathbb{Q})[2] = 0$ (satisfied for 100%) and

$$\alpha = \alpha(2^m, E) = \begin{cases} m/3 & \text{im}(\bar{\rho}_{E,2}) \cong S_3, \\ 2m/3 & \cong \mathbb{Z}/3. \end{cases}$$

For $d = 2$, this improves on Ono from 2000's.

Theorem 15 (Simultaneous nonvanishing). *For 100% of tuples of elliptic curves $E_1, \dots, E_m$ such that $L(E_i, 1) \neq 0$, it holds that: for each d and $\alpha > 0$, we have*

$$\#\{\chi \in F_d(X) : L(E_i, \chi, 1) \neq 0 \forall i\} \gg \frac{X}{(\log X)^{1-\alpha}}.$$

The proofs uses horizontal p -adic L -functions.

Definition 16. Fix a prime p . Let $E/\mathbb{Q}$ be an elliptic curve, of conductor N . We say that ℓ is a *Taylor–Wiles prime* for (E, p) if the following three conditions are met:

- (1) $(\ell, N) = 1$
- (2) $\ell \equiv 1 \pmod{p}$
- (3) $a_E(\ell) \not\equiv 2 \pmod{p}$.

We say that p is *E -good* if Taylor–Wiles primes have positive density.

Remark 17. The last condition (3) is really one concerning the mod p Galois representation $\bar{\rho}_{E,p}$: if the latter is irreducible, then p is E -good.

Remark 18. If E is non-CM and $p \geq 13$, then p is E -good (Zywina).

Thus, let p be E -good. We consider $\mathcal{L} = (\ell_n)_n$, where $\ell_n \equiv 1 \pmod{p}$ and ℓ_n is a Taylor–Wiles prime for all sufficiently large n . Set

$$m_n := v_p(\ell_n - 1) \geq 1.$$

Defint eh *horizontal Iwasawa algebra*

$$\Lambda^{\text{hor}} := \mathbb{Z}_p \left[\left[\prod_{n \in \mathbb{N}} \mathbb{Z}/p^{m_n} \right] \right] := \varprojlim_{\substack{A \subseteq \mathbb{N} \\ \text{finite}}} \cong \text{Hom}_{\text{cts}} \left(\mathcal{C} \left(\prod \mathbb{Z}/p^{m_n}, \mathbb{Z}_p \right), \mathbb{Z}_p \right).$$

Associated to the elliptic curve, we define (using modular symbols) a measure

$$\nu_E \in \Lambda$$

that interpolates the twistsby characters of p -power order, i.e.,

$$\chi : \prod_{n \in \mathbb{N}} (\mathbb{Z}/\ell_n)^* \rightarrow \bar{\mathbb{Q}}^\times.$$

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Now if χ has p -power order, then it factors through some character

$$\tilde{\chi} : \prod_{n \in \mathbb{N}} \mathbb{Z}/p^{m_n} \rightarrow \bar{\mathbb{Q}}^\times.$$

The connection is that

$$\nu_E(\tilde{\chi}) = (\text{Euler factor}) \cdot \frac{L(E, \chi, 1) \tau(\bar{\chi})}{\Omega_E}.$$

Pushing forward along

$$\prod \mathbb{Z}/p^{m_n} \twoheadrightarrow \prod \mathbb{Z}/p = (\mathbb{Z}/p)^\infty,$$

get

$$\overline{\nu_E} \in \mathbb{Z}_p[[(\mathbb{Z}/p)^\infty]] = \Lambda^{\text{diag}}.$$

Theorem 19 (K–N). Let $0 \neq \nu \in \Lambda^{\text{diag}}$. Then $\nu(x) \neq 0$ for a “positive proportion” of characters $\chi : (\mathbb{Z}/p)^\infty \rightarrow \bar{\mathbb{Q}}^\times$. There exists $I \subseteq \mathbb{N}$, a finite set, such that for all $\chi : (\mathbb{Z}/p)^\infty \rightarrow \bar{\mathbb{Q}}^\times$, there exists $\chi_I : \prod_{n \in I} \mathbb{Z}/p \rightarrow \bar{\mathbb{Q}}^\times$ such that $\nu(\chi \chi_I) \neq 0$.

The *upshot* is that it suffices to prove $\overline{\nu_E} \neq 0$. This follows if we can show any of the following:

$$\left\{ \begin{array}{ll} & L(E, 1) = \bar{\nu}_E(1) \neq 0 \\ p = 2 & \text{Friedberg-Hoffstein,} \\ \text{general } p & \text{use Kurihara's conjecture.} \end{array} \right.$$

REFERENCES